Computational Statistics and Machine Learning project

Kickoff meeting

Prof. Dr. Seth Flaxman

RPTU Kaiserslautern-Landau

Winter term 2026/27
Wednesday 28.10.2026, Building 48, Room 379

Course website

https://mlgh.net/rptu-csml-2026

Class repo

https://github.com/MLGlobalHealth/rptu-csml-2026

To do: get a GitHub account and fork the repo

Deadline: Friday 30.10.2026

Researchers who will be helping advise projects



Ahmed Abdelrahman

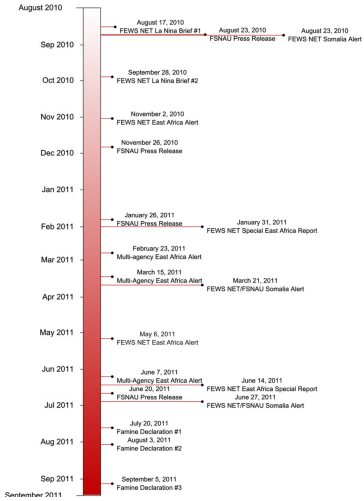


Sahoko Ishida

Week 1 readings

- Hillbruner, C. and Moloney, G. (2012). When early warning is not enough: Lessons learned from the 2011 Somalia famine. *Global Food Security* 1(1), 20–28.
- Murphy, A. H. (1993). What is a good forecast? An essay on the nature of goodness in weather forecasting. *Weather and Forecasting* 8(2), 281–293.

Early warning products, Aug 2010 – Sep 2011



Hillbruner & Moloney (2012), Fig. 1: special early warning products produced by FEWS NET and FSNAU.

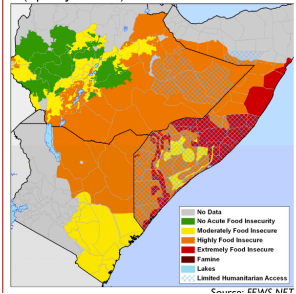
EAST AFRICA Food Security Alert — March 15, 2011

Below-average March to May rains forecast in the Eastern Horn – current crisis likely to worsen

In the eastern Horn of Africa, households in pastoral and marginal cropping areas currently face moderate to extreme levels of food insecurity due to an ongoing drought, deteriorating purchasing power, and, in some areas, limits on the delivery of humanitarian assistance. Based on the findings of a multi-agency scenario building process, the likely poor performance of March-May rainfall is expected to result in further deterioration in food security (Figures 1 and 2). Even if March-May rains are average, food security in the region's eastern sector is expected to remain critical through at least May/June 2011. Large-scale emergency assistance to address current and likely food insecurity is needed; additional contingency planning, given the possibility of a major crisis, should also be implemented.

Following two seasons of average to above-average rainfall in most areas, the 2010 October-December rains were extremely poor – by some measures the worst in thirty years – across the region's eastern sector. January/February harvests completely failed in most cropping areas. In pastoral areas, poor rainfall has severely limited both livestock conceptions and pasture/water availability. Though

Figure 1. Most-likely food security outcomes (April - June 2011)



Source: FEWS NET

FSNAU, 15 March 2011: worst case scenario

In the worst case scenario, rains would be less than 50 percent of average. Substantial crop failure and massive livestock mortality would occur, resulting in an expansion of extreme food insecurity across much of the region. **In marginal cropping areas of Juba and Hiran (southern Somalia), where humanitarian access is constrained, and the median GAM prevalence had already exceeded 25 percent as of December 2010, localized famine conditions, including significantly increased child mortality, are possible if the worst case scenario assumptions are realized.** Even if March-May rains are average, food security in the region is not expected to improve until at least May/June 2011.

FEWS NET / FSNAU, 20 July 2011

FSNAU

FEWS NET
FAMINE EARLY WARNING SYSTEMS NETWORK

FAMINE IN SOUTHERN SOMALIA

Evidence for a declaration

Nairobi, Kenya
July 19, 2011

















FSNAU

FEWS NET
FAMINE EARLY WARNING SYSTEMS NETWORK

Key Messages

1. Evidence indicates that famine currently exists in Bakool agropastoral and Lower Shabelle and that a humanitarian emergency exists across the rest of the south.

2. This crisis represents the most serious food insecurity situation in the world today, in terms of both scale and severity. Current humanitarian response is inadequate to meet emergency needs. Assuming current levels of response, evidence suggests that famine across all regions of the south will occur in the coming 1-2 months.

3. A massive multisectoral response is critical to prevent additional deaths and total livelihood/social collapse. Most immediately, interventions to improve food access and to address health/nutrition issues are needed. In the medium term, interventions to rebuild and support livelihoods are critical. Extraordinary measures to provide this response should be implemented.

4. Tens of thousands of lives can be saved, but the window of opportunity to do so is extremely limited. These assistance needs will persist through at least December 2011.

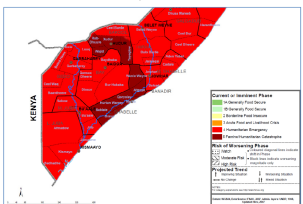
FEWS NET / FSNAU, 20 July 2011: current and projected

FSNAU

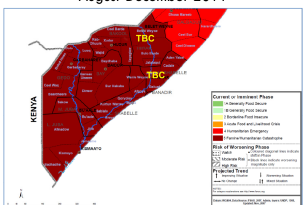
FEWS NET
FAMINE EARLY WARNING SYSTEMS NETWORK

Key Messages

Current food security classification, July 2011

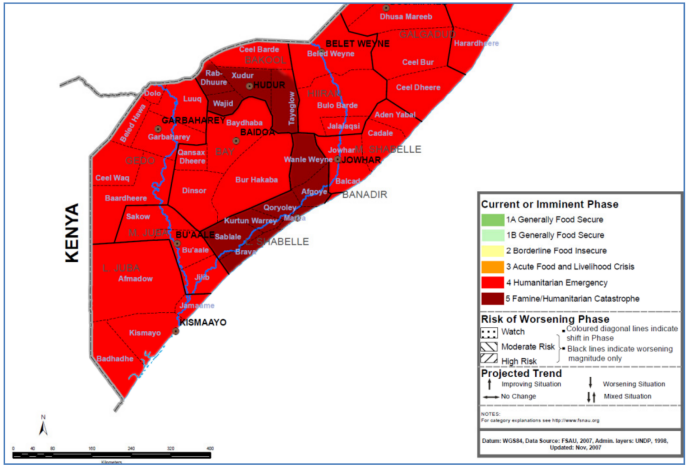


Most-likely food security classification, August-December 2011

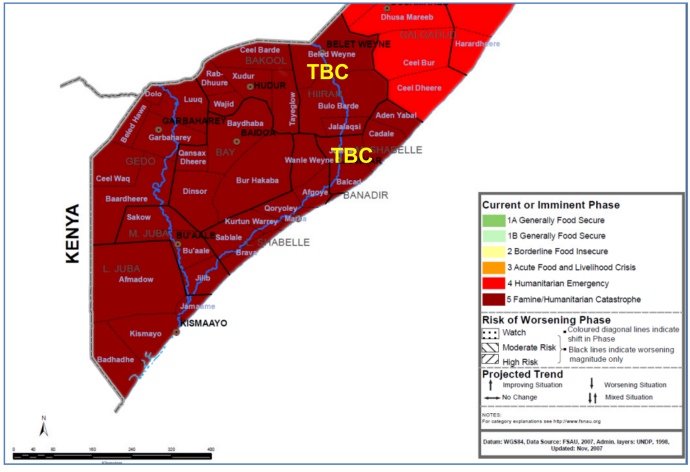


TBC – Final classification awaiting nutrition and mortality survey data which is expected July 27th

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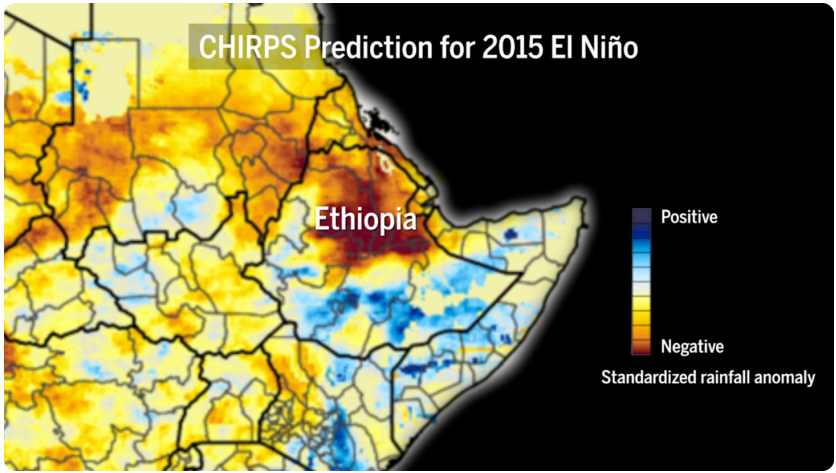


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Science: How early warnings of drought help prevent famines



Science Magazine (4:36)
https://www.youtube.com/watch?v=ddCCENYOIgM

Al Jazeera, *Fault Lines*, 28 November 2011



Horn of Africa Crisis: Somalia's Famine (24:49)
https://www.youtube.com/watch?v=iRZPMd1H7Z0

Outline

Intro

Supervised vs unsupervised learning

R and RStudio

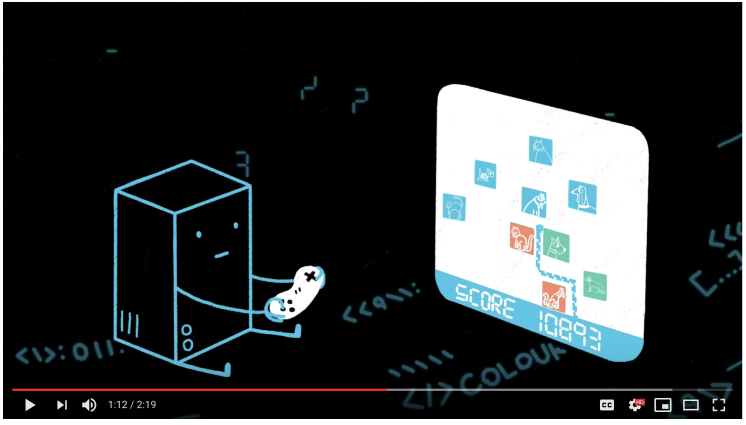
Supervised learning

Linear models

Loss functions

Optimization

What is machine learning?



https://www.youtube.com/watch?v=f_uwKZIAeM0

What is machine learning?

For our purposes, follow Tom Mitchell:

> *A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E.*

What is statistical machine learning?

Both provide methods for learning from data.

Computer science takes an algorithmic perspective: propose an algorithm for data, study the algorithm formally.

Statistics takes an inferential perspective: propose a model for data, study the model formally.

Statistical machine learning (and computational statistics) is the intersection: algorithmic perspective on statistical methods, statistical perspective on algorithms.

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Loss functions

$L(x, y, f)$: given input x and output y, evaluate function f.

$$\text{Does } f(x) \approx y?$$

- Loss function: standard choice in regression is squared error (L_2) loss:
$$L(x, y, f) := (y - f(x))^2 \tag{5}$$
- Standard choice in classification is misclassification rate (1 - accuracy):
$$L(x, y, f) := \mathcal{I}(y \neq f(x)) \tag{6}$$

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Loss is bad, you want to avoid loss, so smaller loss is better! (Some losses are always positive, others can be positive or negative.)

Loss functions: statistical perspective

- Most ML loss functions have a probabilistic perspective.
- All probabilistic likelihoods can be used as losses. How? (remember: avoid loss!)

Loss functions: statistical perspective

- Most ML loss functions have a probabilistic perspective.
- All probabilistic likelihoods can be used as losses. How? (remember: avoid loss!)
- Loss = negative log-likelihood

Project ideas

1. Retrospective reevaluation of FEWS NET
2. Infrastructure to benchmark our famine forecasts
3. Forecasting with open data (climate, price data)
4. Forecasting with NLP data (news sources, social media)
5. Forecasting with satellite imagery
6. Uncertainty quantification
7. A project you design...

Office hours

Office hours once per week

Time and date will be posted on the course website:

https://mlgh.net/rptu-csml-2026